



Escuela Técnica Superior de
Ingeniería Informática

FUNDAMENTOS DE INGENIERÍA DE DATOS

Estudio de la depresión estudiantil

Realizado por:

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Jaime Linares Barrera
Daniel Vela Camacho

Grupo 5

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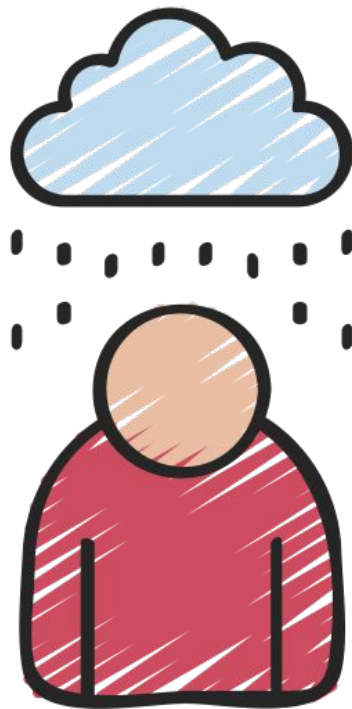
CONCLUSIONES

01

INTRODUCCIÓN



- PROBLEMA **COMÚN**
- **DIFÍCIL** DE TRATAR
- CONSECUENCIAS **FATALES**



01

PREGUNTAS PLANTEADAS



¿QUÉ FACTORES **PROVOCAN** DEPRESIÓN EN LOS **ESTUDIANTES**?

¿**INFLUYE** EL **SEXO** O EL **NIVEL ACADÉMICO**?

¿PODEMOS **PREDECIR** LA DEPRESIÓN?

02

DATOS



- 18 ATRIBUTOS
- 27902 FILAS
- 56% HOMBRES Y 44% MUJERES
- PORCENTAJE DE DEPRESIÓN; 41% NO Y 58% SÍ





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494



Code



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Student Depression Dataset

Analyzing Mental Health Trends and Predictors Among Students

[Data Card](#)[Code \(102\)](#)[Discussion \(5\)](#)[Suggestions \(0\)](#)

About Dataset

Student Depression Dataset: Analyzing Mental Health Trends and Predictors Among Students

Overview

This dataset compiles a wide range of information aimed at understanding, analyzing, and predicting depression levels among students. It is designed for research in psychology, data science, and education, providing insights into factors that contribute to student mental health challenges and aiding in the design of early intervention strategies.

Data Description

Usability ⓘ

10.00

License

[Apache 2.0](#)

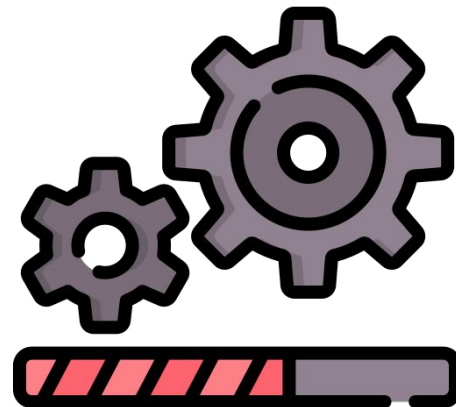
Expected update frequency

Annually

Tags

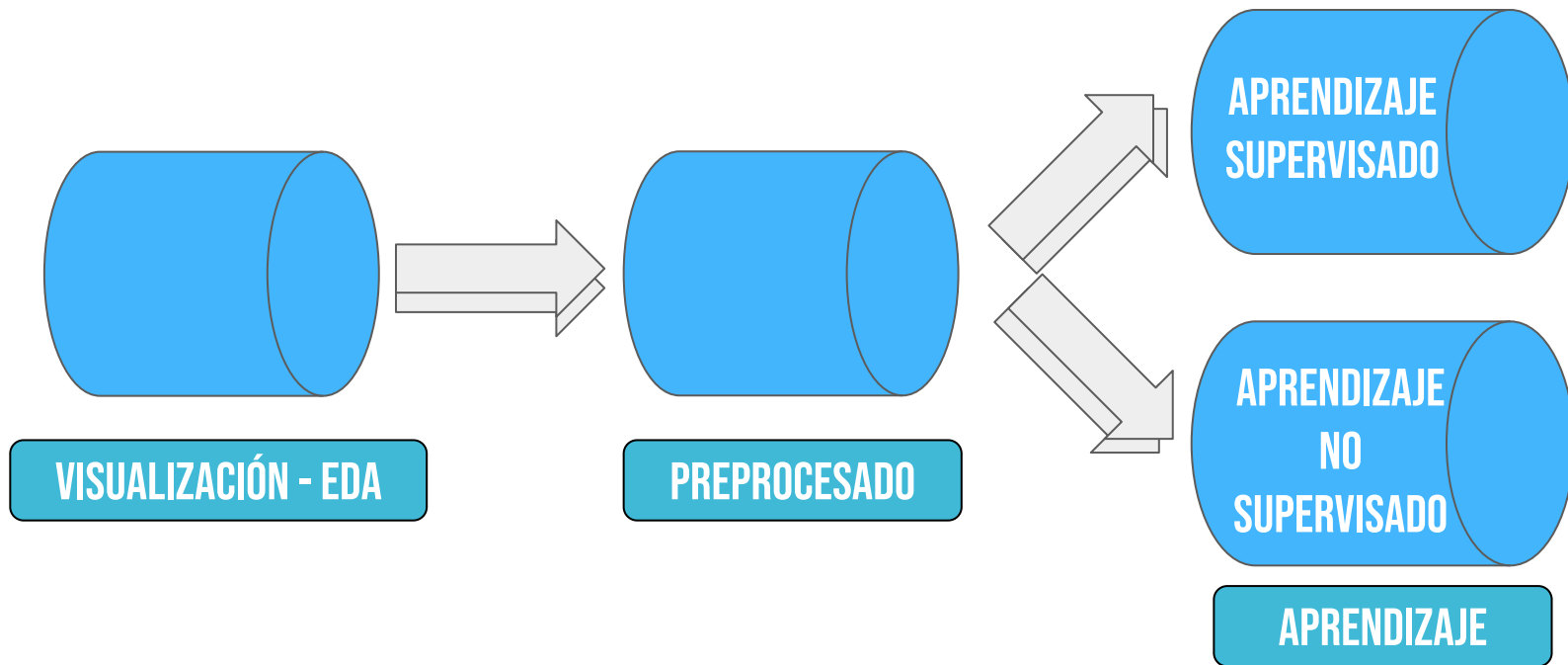
MÉTODOS UTILIZADOS

- SELECCIÓN DE ATRIBUTOS Y EJEMPLARES
- ALGORITMOS DE CLASIFICACIÓN
- SHAP
- USO DE K-MEDIoidES
- REGLAS DE ASOCIACIÓN



03

WORKFLOW

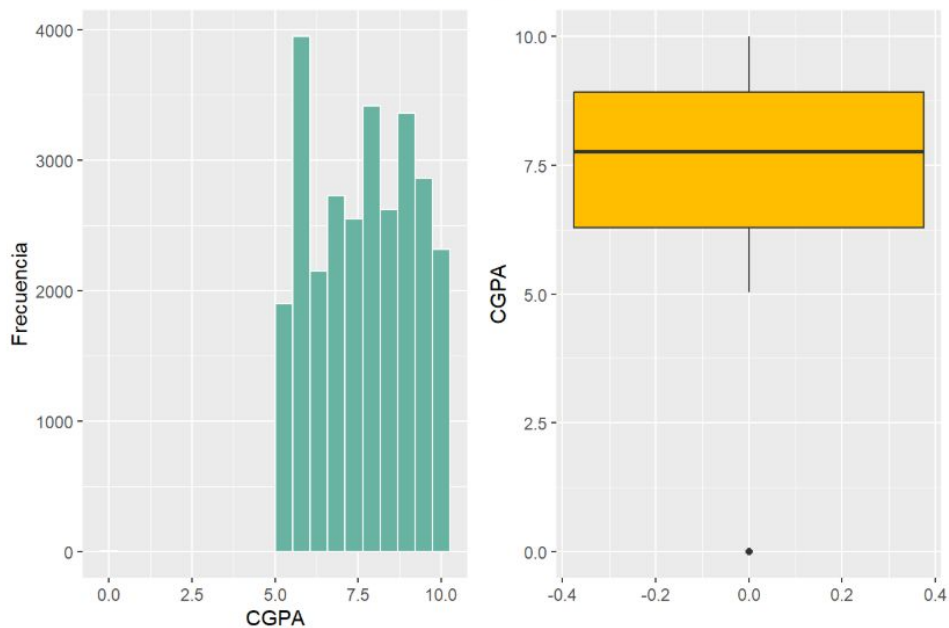


04

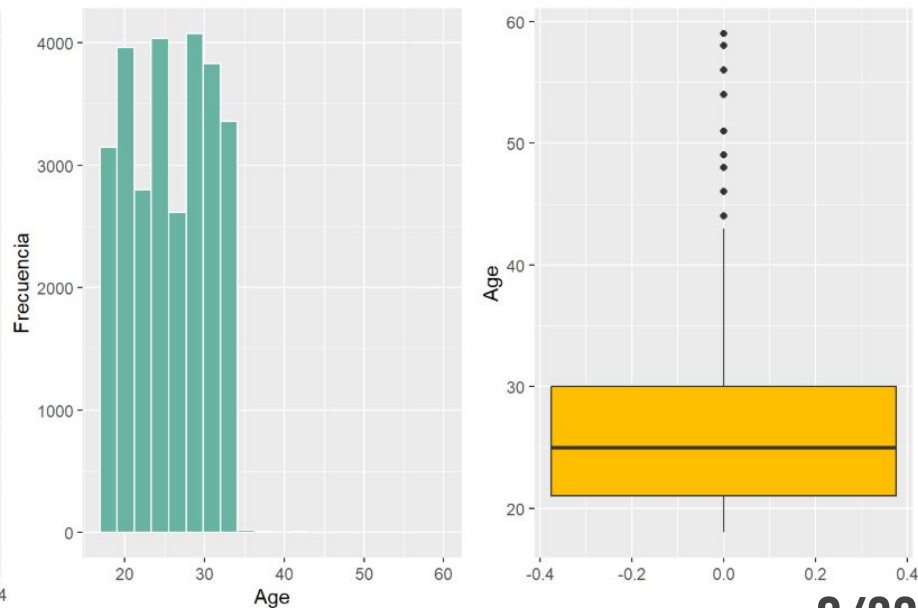
RESULTADOS VISUALIZACIÓN - EDA



Visualización univariante de CGPA



Visualización univariante de Age

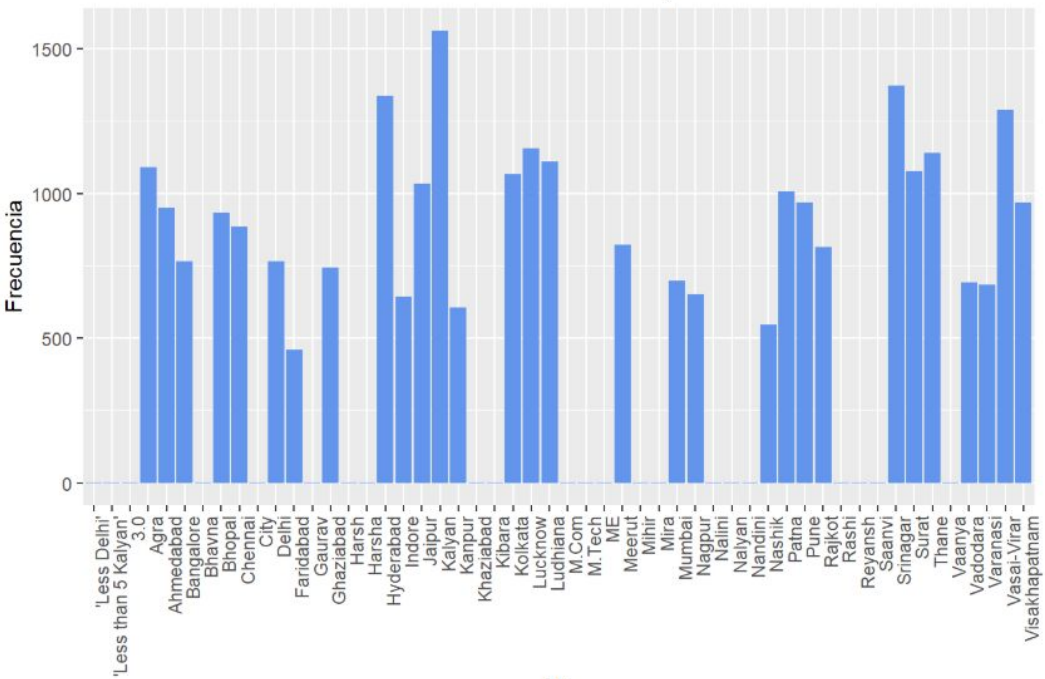


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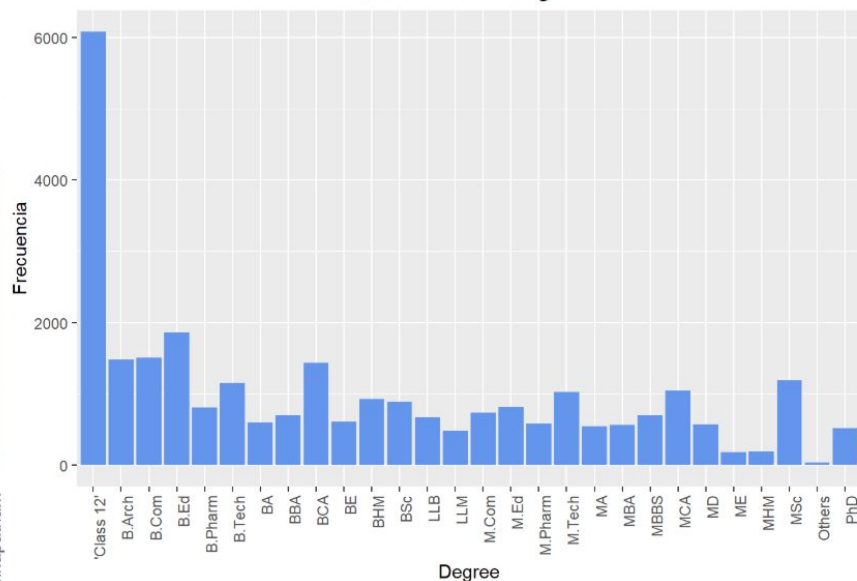
RESULTADOS VISUALIZACIÓN - EDA



Distribución de City

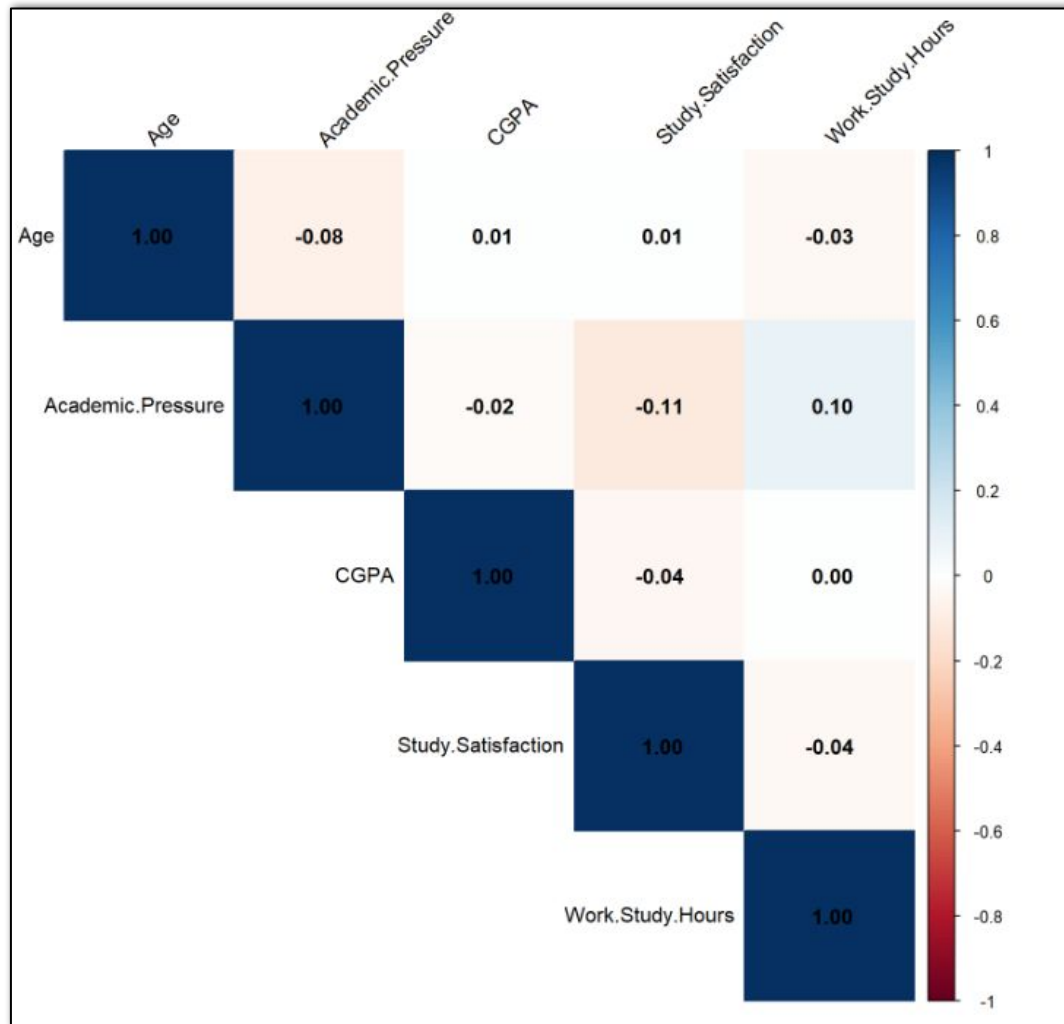


Distribución de Degree



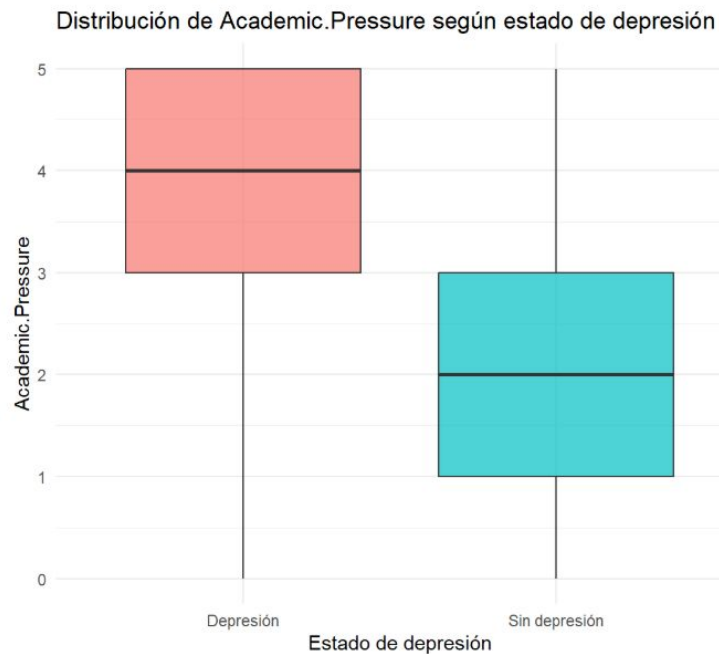
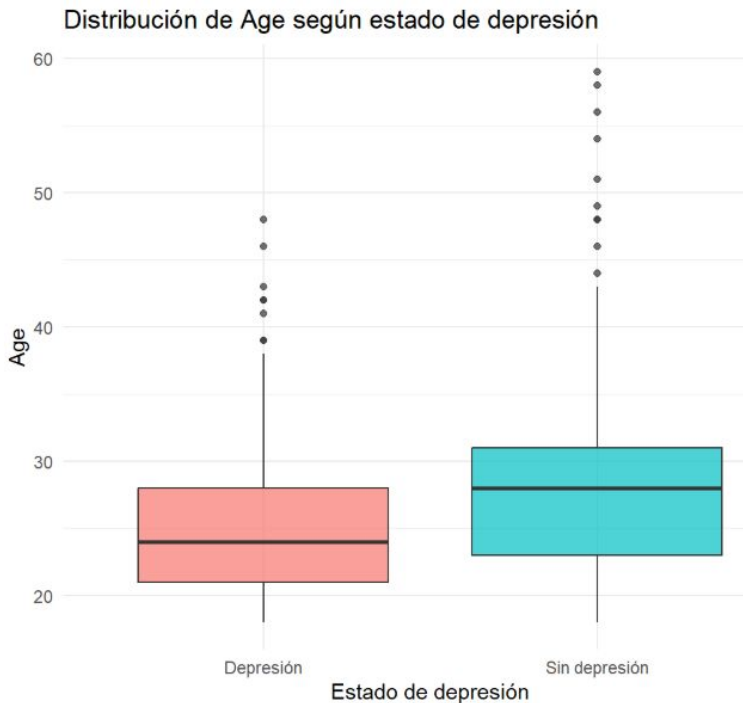
03

RESULTADOS



04

RESULTADOS VISUALIZACIÓN - EDA



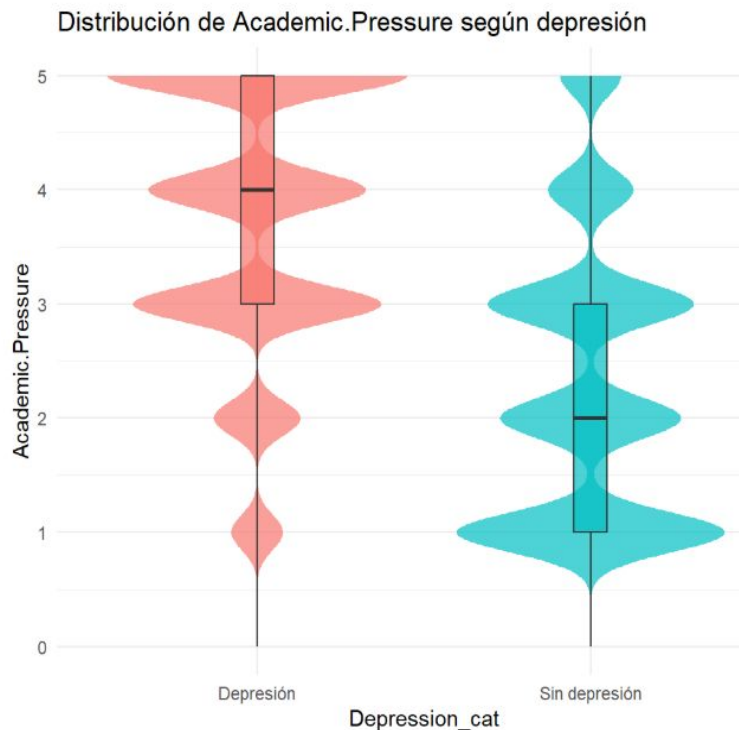
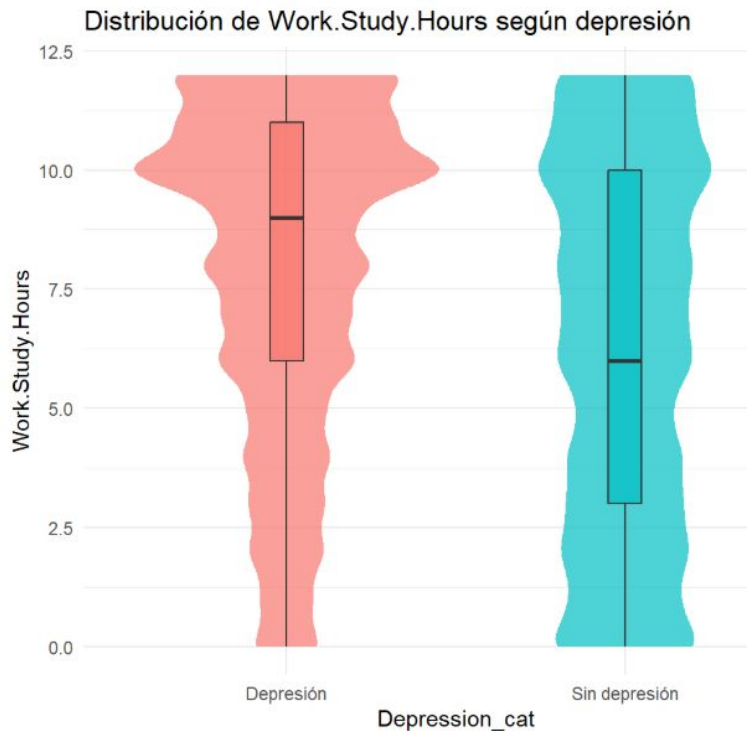
Depresión

Depresión

Sin depresión

04

RESULTADOS VISUALIZACIÓN - EDA

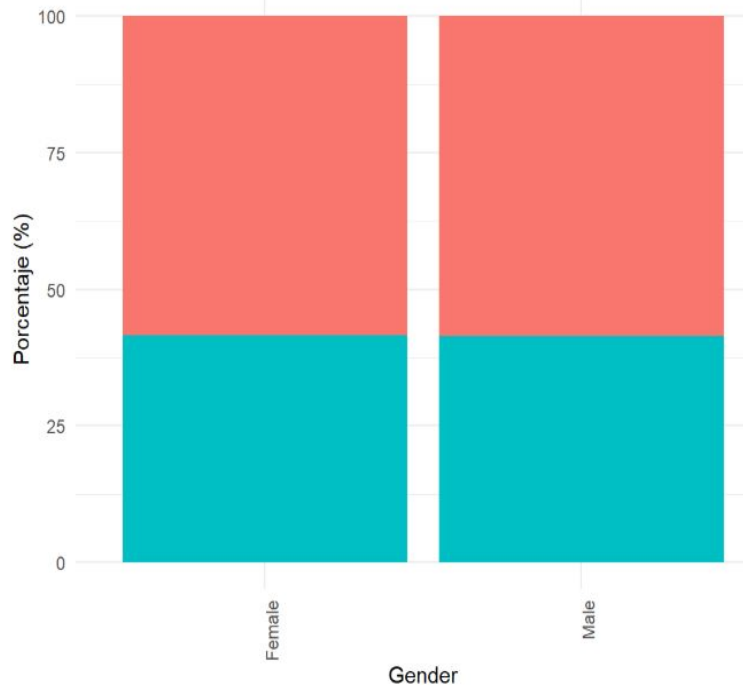


Depresión

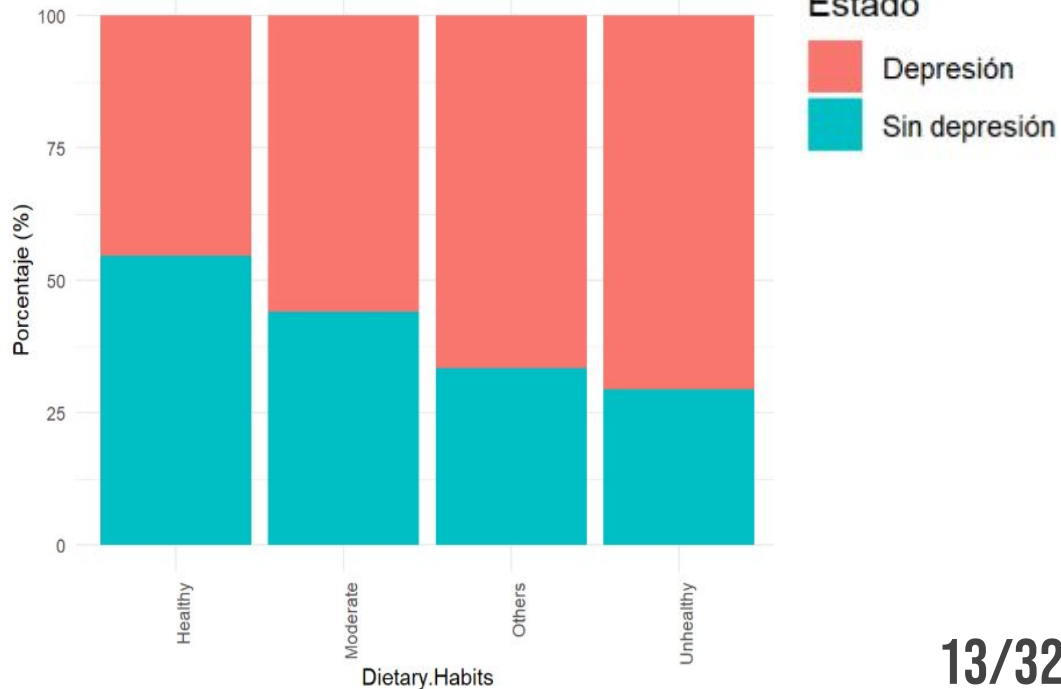
Depresión

Sin depresión

Porcentaje de estudiantes deprimidos según Gender



Porcentaje de estudiantes deprimidos según Dietary.Habits



04

RESULTADOS PREPROCESADO



TRATAMIENTO DE DATOS

ELIMINACIÓN DE ATRIBUTOS

LIMPIEZA DE VALORES **INCONSISTENTES**

TRATAMIENTO DE ATRIBUTOS **ORDINALES**

REDUCCIÓN CARDINALIDAD ATRIBUTOS ORDINALES

GESTIÓN DE **OUTLIERS**

04

RESULTADOS PREPROCESADO

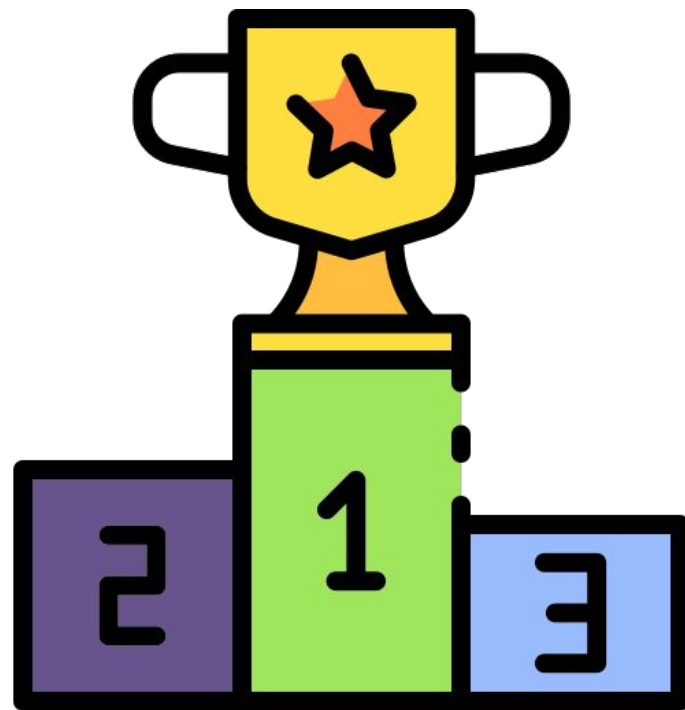


VARIABLES MÁS INFORMATIVAS (MI)

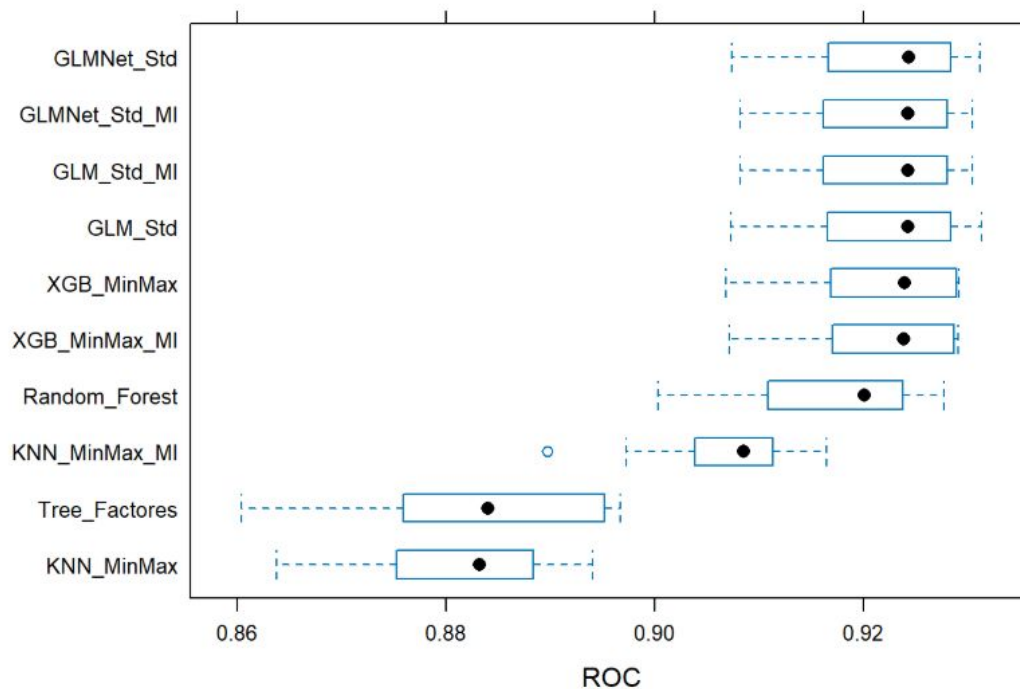
PENSAMIENTOS SUICIDAS

PRESIÓN ACADÉMICA

ESTRÉS FINANCIERO



Comparación de modelos (ROC en CV)

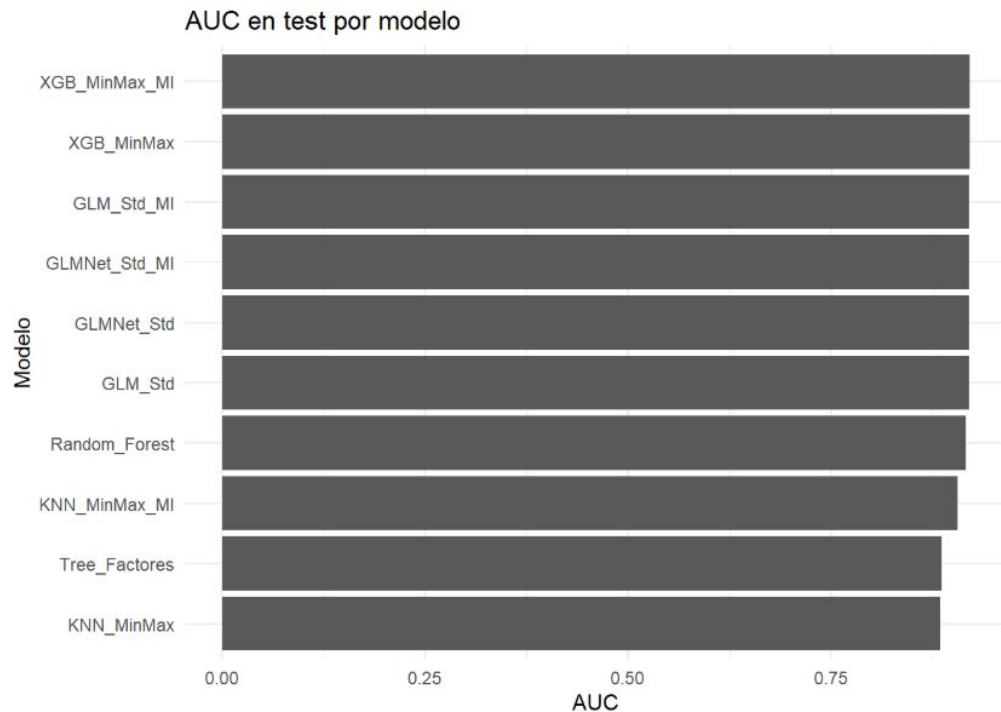


04

RESULTADOS APRENDIZAJE SUPERVISADO



XGB_MINMAX_MI

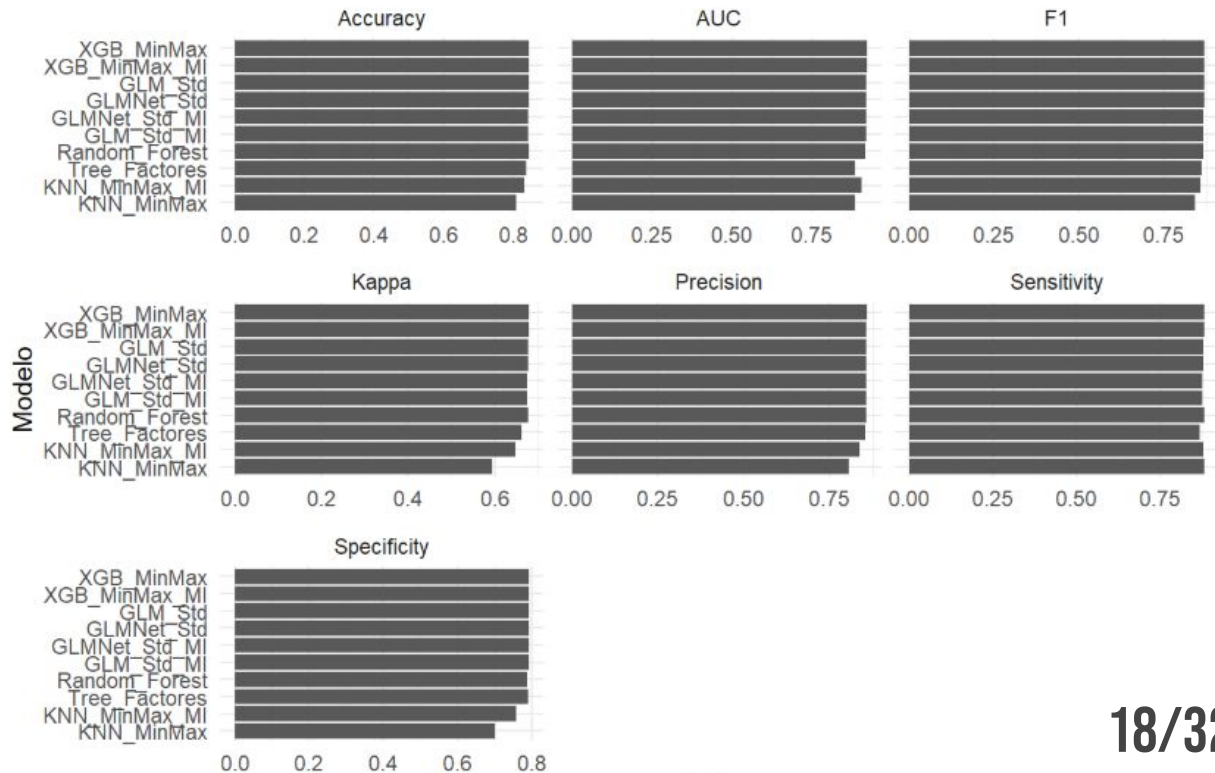


04

RESULTADOS APRENDIZAJE SUPERVISADO



OTRAS MÉTRICAS

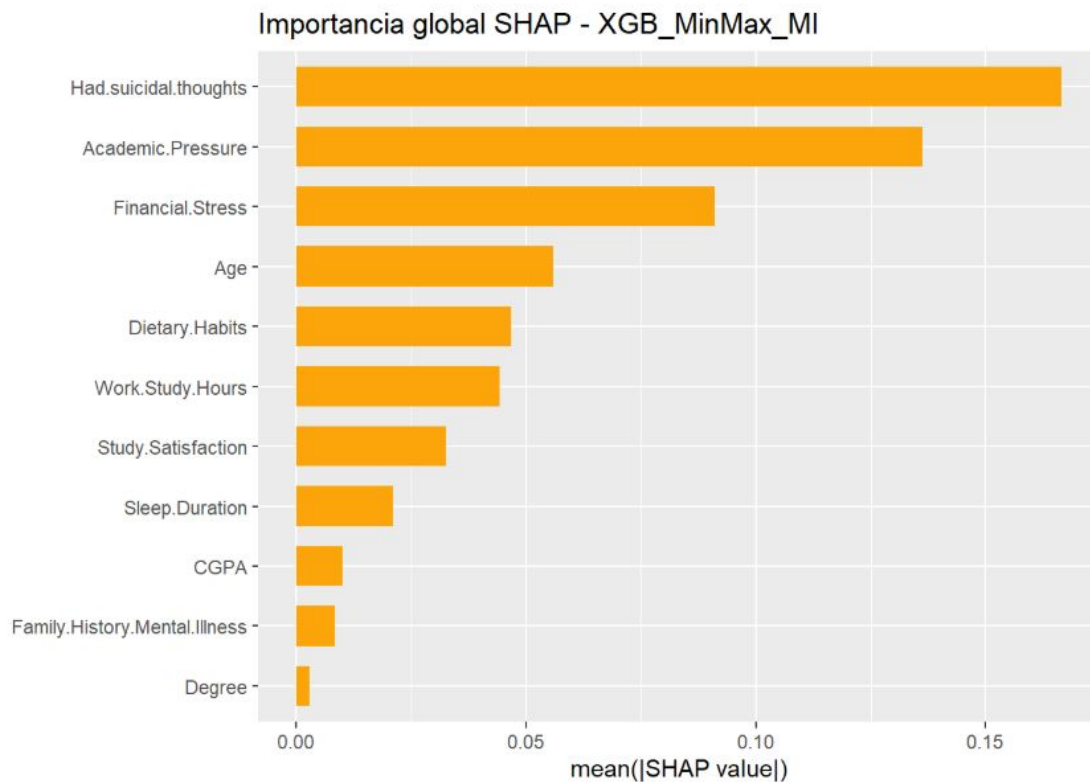


04

RESULTADOS APRENDIZAJE SUPERVISADO



SHAP



04

RESULTADOS APRENDIZAJE SUPERVISADO



SHAP



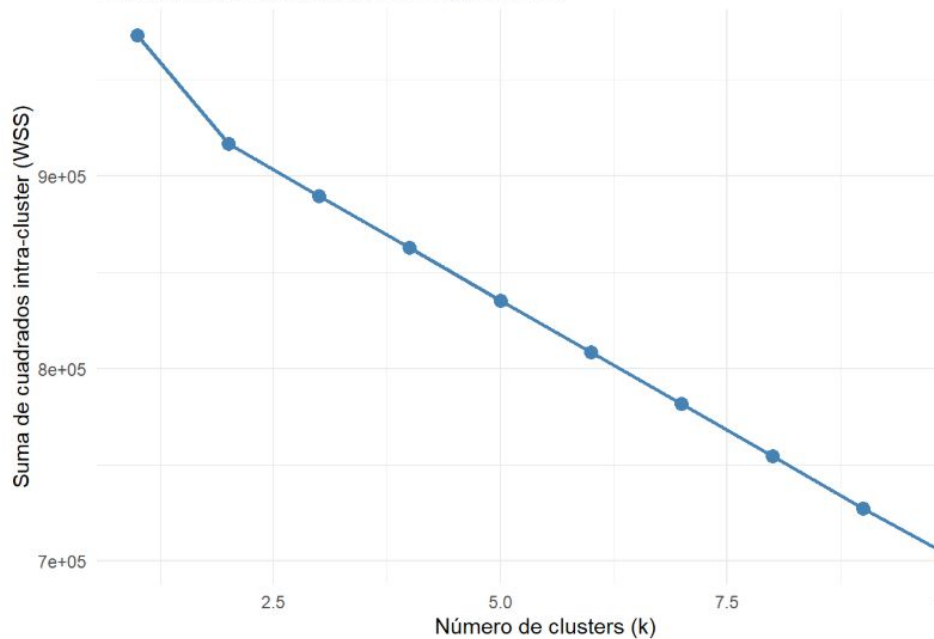
04

RESULTADOS APRENDIZAJE NO SUPERVISADO



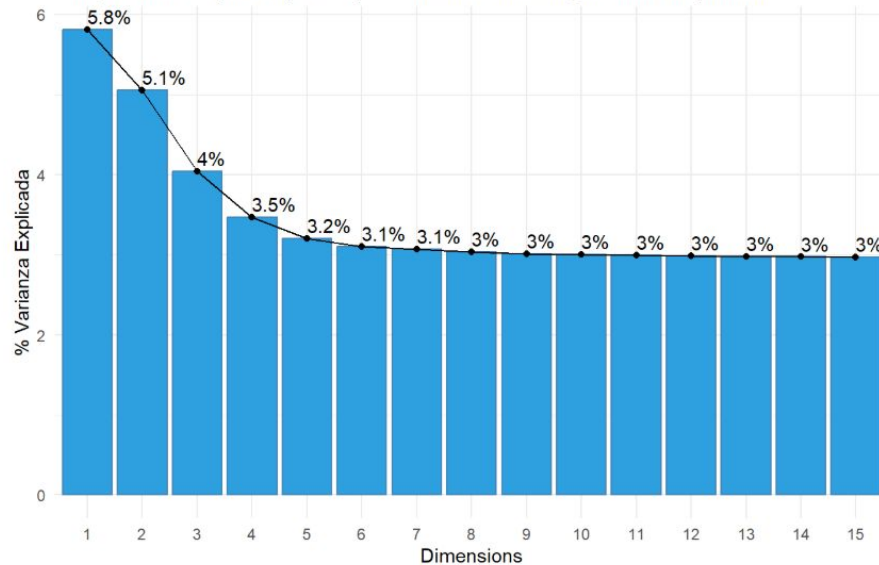
Método del Codo (WSS)

Cálculo iterativo manual (MacQueen | nstart=20)



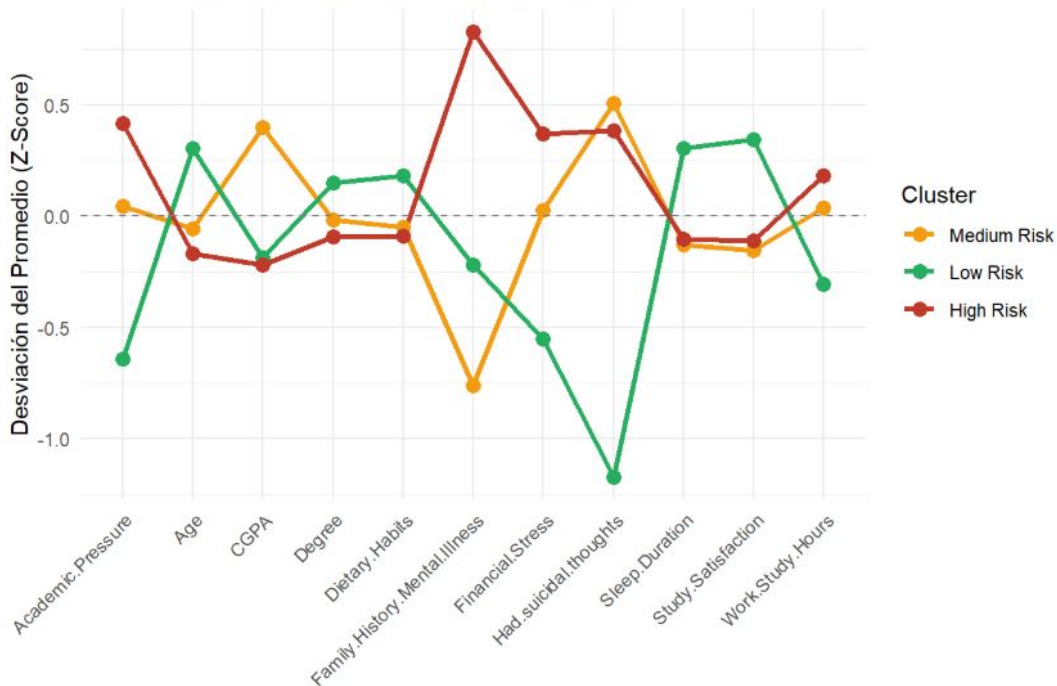
Sedimentación de Varianza (Dataset Completo)

La baja varianza explicada por componentes individuales sugiere alta entropía/ruído

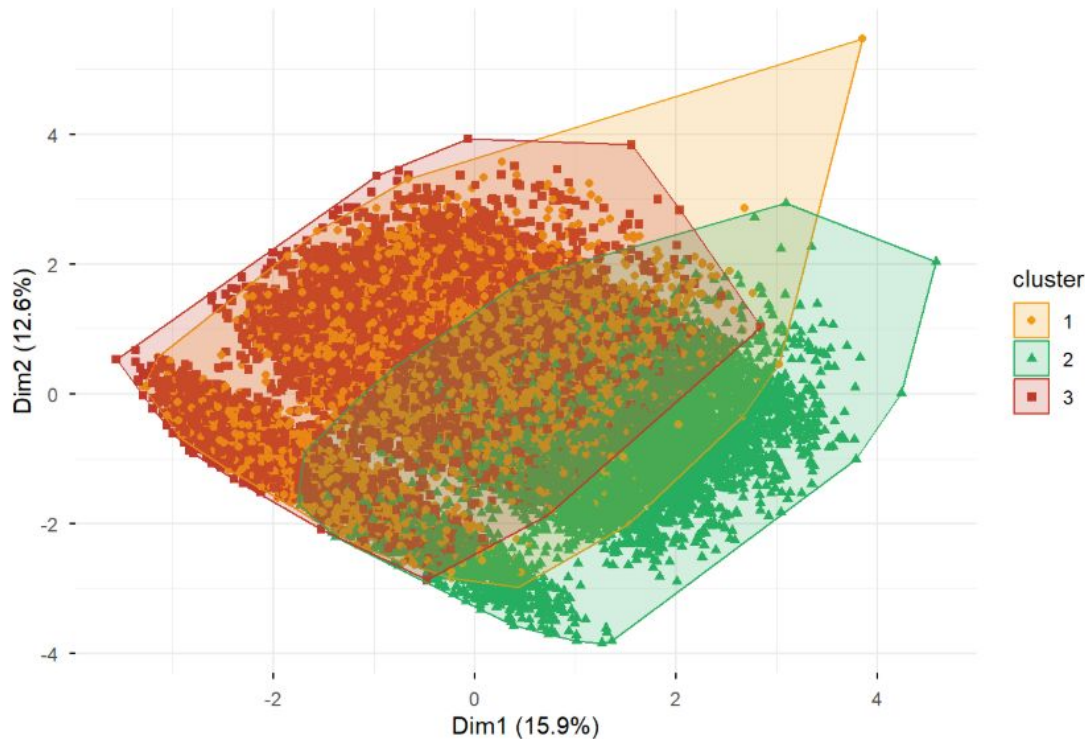


Perfil Clínico de los Clusters (Snake Plot)

Z-Scores Promedio: Identificación de perfiles de riesgo

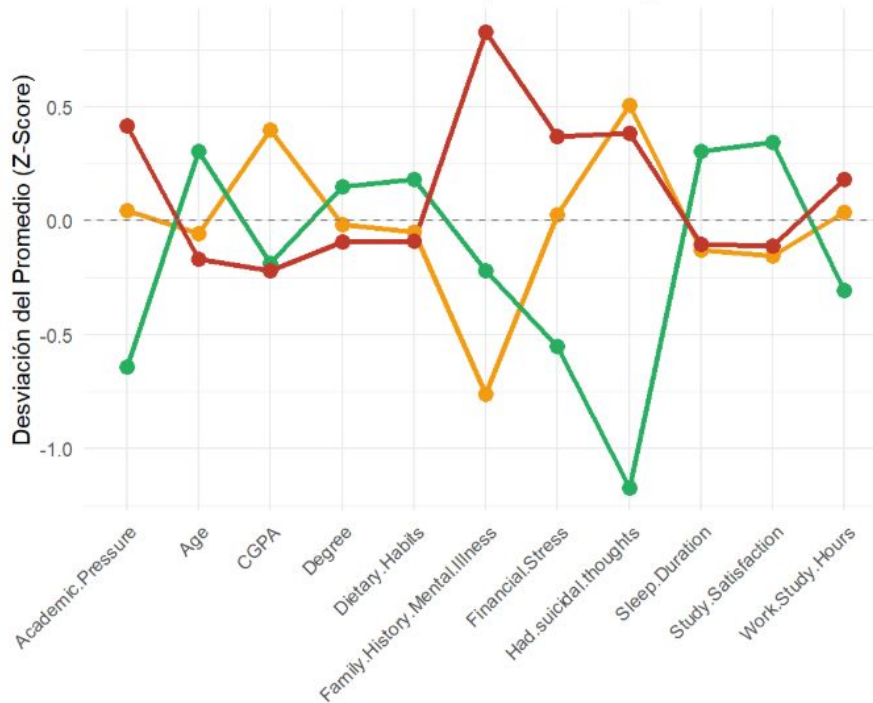


Cluster Plot: K-Medoids (CLARA, k=3)



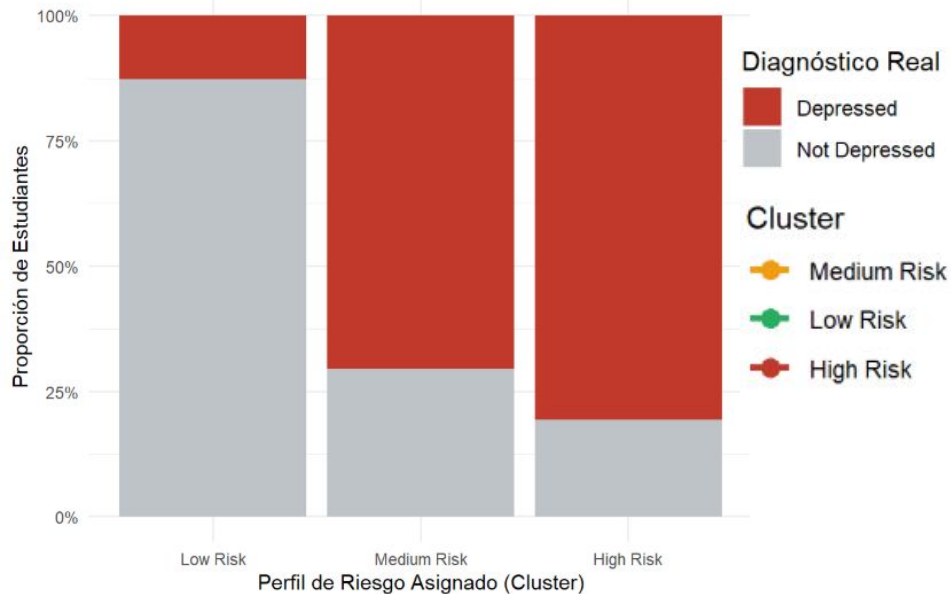
Perfil Clínico de los Clusters (Snake Plot)

Z-Scores Promedio: Identificación de perfiles de riesgo



Validación Externa: Prevalencia de Depresión por Perfil

¿Coincide el riesgo calculado (Cluster) con el diagnóstico real?



PRESIÓN ACADÉMICA **EXTREMA** + ESTRÉS FINANCIERO **CRÍTICO** + PENSAMIENTOS **SUICIDAS**

Top 10 Patrones de Riesgo Identificados

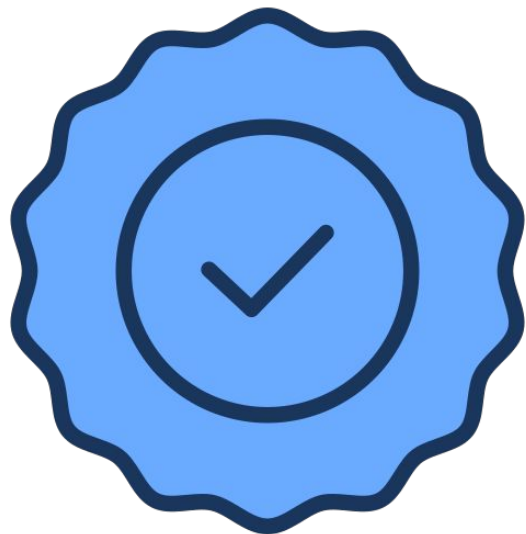
Regla (Causa → Efecto)	Soporte	Confianza	Lift	Conteo
Academic_Pressure=Very High,Financial_Stress=Critical,Suicidal_Thoughts=Yes → Depression	0.056	0.984	1.681	1553
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Dietary_Habits=Unhealthy,Work_Load=High → Depression	0.050	0.974	1.665	1403
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Dietary_Habits=Unhealthy → Depression	0.073	0.968	1.655	2020

04

RESULTADOS APRENDIZAJE NO SUPERVISADO



Regla (Causa → Efecto)	Soporte	Confianza	Lift	Conteo
Academic_Pressure=Very High,Financial_Stress=Critical,Suicidal_Thoughts=Yes → Depression	0.056	0.984	1.681	1553
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Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Dietary_Habits=Unhealthy → Depression	0.073	0.968	1.655	2020
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Family_History=Yes,Work_Load=High → Depression	0.057	0.968	1.654	1589
Financial_Stress=Critical,Suicidal_Thoughts=Yes,Dietary_Habits=Unhealthy,Work_Load=High → Depression	0.053	0.963	1.645	1478
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Gender=Male,Work_Load=High → Depression	0.060	0.959	1.639	1677
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Work_Load=High → Depression	0.112	0.959	1.639	3115
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Sleep_Duration=Less than 5 hours → Depression	0.057	0.959	1.639	1580
Academic_Pressure=Very High,Financial_Stress=Critical → Depression	0.066	0.954	1.630	1825
Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Family_History=Yes → Depression	0.083	0.952	1.626	2316



VALIDACIÓN EXTENSA

EL MEJOR MODELO **XGB_MINMAX_MI**
FACTORES **DETERMINANTES** DETECTADOS
FACTORES **AMPLIFICADORES**

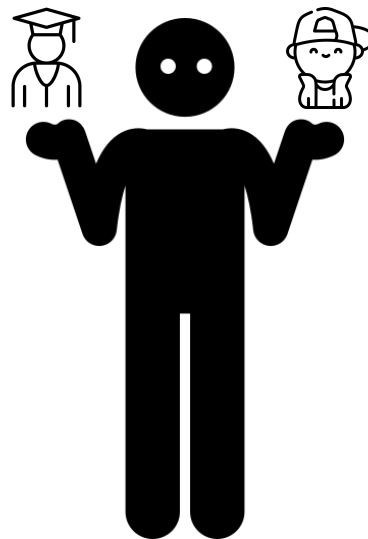
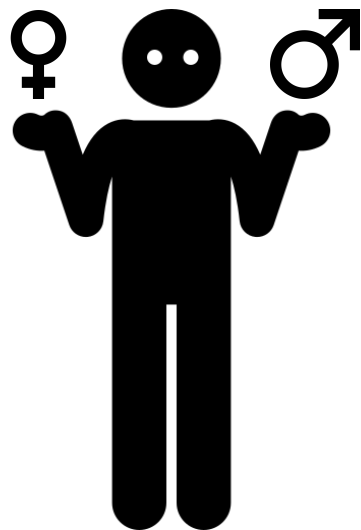
¿QUÉ FACTORES **PROVOCAN** DEPRESIÓN EN LOS **ESTUDIANTES**?

PENSAMIENTOS SUICIDAS

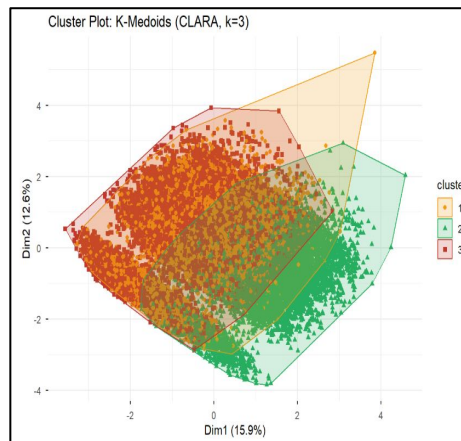
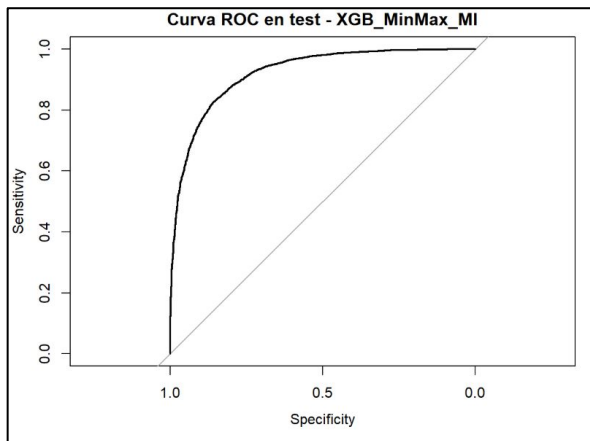
PRESIÓN ACADÉMICA

ESTRÉS FINANCIERO

¿INFLUYE EL SEXO O EL NIVEL ACADÉMICO?



¿PODEMOS PREDECIR LA DEPRESIÓN?



Top 10 Patrones de Riesgo Identificados

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Academic_Pressure=Very High,Finacial_Stress=Critical,Suicidal_Thoughts=Yes → Depression	0.056	0.984	1.681	1553
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Academic_Pressure=Very High,Suicidal_Thoughts=Yes,Family_History=Yes → Depression	0.083	0.952	1.626	2316

Lift > 1 indica una asociación positiva fuerte (mayor riesgo).



GITHUB

MUCHAS GRACIAS



DATASET

CRÉDITOS: Esta presentación incluye iconos de [Flaticon](#) e [Icons Icons](#)

Fuentes y colores utilizados

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Bebas Neue

(<https://fonts.google.com/specimen/Bebas+Neue>)

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#40B9D7